Euclid, Book I, Proposition I.7

On the Same Side of a Given Base There Cannot Exist Two Distinct Vertices Determining Equal Corresponding Sides

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1 Introduction

Proposition I.7 occupies a special position in Book I of Euclid's *Elements*. Unlike the preceding propositions, its principal purpose is not to establish a new geometric construction or a direct property of triangles, but to prove a uniqueness principle.

Euclid shows that, on the same side of a given base, there cannot exist two distinct vertices determining triangles whose corresponding sides are equal. This uniqueness result immediately precedes Proposition I.8, where it is used as a crucial ingredient in the classical proof of the Side–Side–Side congruence theorem.

The Hilbert reconstruction follows a different logical order. The general Side–Side–Side theorem has already been established within the Hilbert interface. Consequently, Proposition I.7 becomes a concise uniqueness argument derived from the developed congruence theory, without reproducing Euclid's original chain of auxiliary arguments.

This proposition therefore provides one of the clearest examples of the difference between the historical organization of Euclid's *Elements* and the logical organization of a modern axiomatic development.

1.1 Initial Configuration

Let AB be a given finite straight line and let C and D be two points lying on the same side of the line AB.

Assume that

$$AC \cong AD$$

and

$$BC \cong BD.$$

The proposition asks whether the two vertices C and D can be distinct under these assumptions.

Euclid's objective is to prove that such a configuration is impossible.

1.2 The Assumed Second Vertex

Suppose that the two vertices are distinct.

Both triangles share the same base AB, and their corresponding sides are assumed to satisfy

$$AC \cong AD, \qquad BC \cong BD.$$

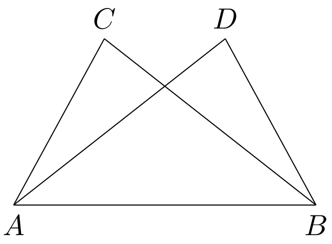
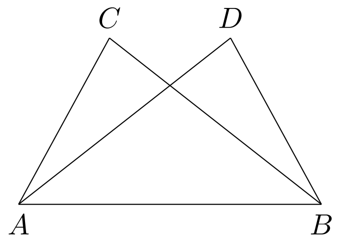


Figure 1: The assumed existence of two distinct vertices on the same side of the base AB.

Thus two different triangles would have the same prescribed side lengths while lying on the same side of the base.

Euclid proceeds by showing that this assumption inevitably leads to a contradiction.



$$AC \cong AD, \qquad BC \cong BD$$

Figure 2: The two assumed triangles have the same base and equal corresponding sides.

1.3 The Contradictory Configuration

Euclid joins the two assumed vertices by the segment CD.

Since

$$AC \cong AD,$$

the triangle ACD is isosceles. By Proposition I.5,

$$\angle ACD \cong \angle CDA.$$

From the geometric position of the points, one of these angles is strictly greater than the corresponding angle formed with the base vertex B.

On the other hand, since

$$BC \cong BD,$$

the triangle BCD is also isosceles. Proposition I.5 therefore gives

$$\angle BCD \cong \angle CDB.$$

The same pair of angles is thus forced to be both equal and unequal, which is impossible. Hence the two assumed vertices cannot be distinct.

2 Statement

Theorem 1 (Euclid I.7)**.** *Let AB be a given base, and let C and D lie on the same side of the line AB.*

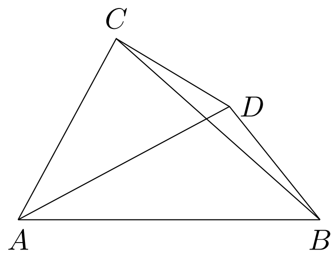


Figure 3: Euclid joins the two assumed vertices C and D and applies Proposition I.5 to the isosceles triangles ACD and BCD.

Assume

$$AC \cong AD$$

and

$$BC \cong BD.$$

Then

$$C = D.$$

Equivalently, on a fixed side of a given base there is at most one point having prescribed distances from the two endpoints of that base.

3 Proof

Assume, for contradiction, that C and D are distinct.

Join C to D. Since

$$AC \cong AD,$$

the triangle ACD is isosceles. By Proposition I.5, its base angles are congruent:

$$\angle ACD \cong \angle CDA.$$

Similarly, since

$$BC \cong BD,$$

the triangle BCD is isosceles, and Proposition I.5 gives

$$\angle BCD \cong \angle CDB.$$

The position of the points on the same side of the base forces one of these angles to contain the other as a proper part. Hence an angle would have to be congruent to one of its proper parts, which is impossible.

Therefore the assumption that C and D are distinct leads to a contradiction. Hence

$$C = D.$$

Thus, on a fixed side of a given base, a point is uniquely determined by its distances from the two endpoints of the base.

4 Implementation Architecture

The formal proof bears little resemblance to Euclid's original argument.

Instead of constructing auxiliary triangles and reasoning by contradiction, the implementation first applies the already established Side–Side–Side theorem of the Hilbert interface. This immediately shows that the two candidate triangles are congruent.

From the resulting congruence, the corresponding angle at the common base vertex is obtained. The uniqueness of angle construction then forces both candidate vertices to lie on the same ray issued from the base vertex.

Finally, since the two points lie on the same ray and are at the same distance from the common endpoint, the uniqueness of segment construction implies that the two points coincide.

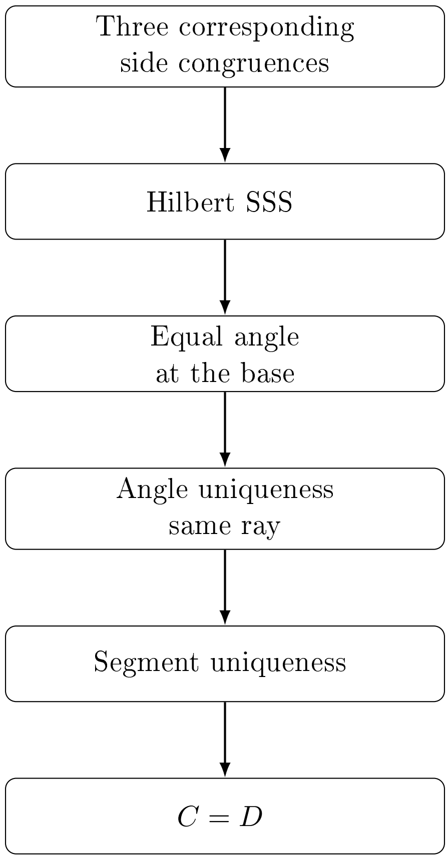


Figure 4: Architecture of the Hilbert reconstruction of Proposition I.7.

5 Hilbert Reconstruction

The Hilbert reconstruction follows a completely different line of reasoning from Euclid's original proof.

The three assumed side congruences are first combined with the nondegeneracy of one of the triangles. The previously established Hilbert Side–Side–Side theorem immediately yields congruence of the two triangles and, in particular, equality of the corresponding angles at the common base vertex.

Since both candidate vertices lie on the same side of the base, the uniqueness theorem for angle construction implies that they determine the same ray issuing from the base endpoint.

Finally, the two points lie on the same ray and are at the same distance from the common endpoint. The uniqueness of segment construction therefore forces the two points to coincide.

Thus the uniqueness statement of Proposition I.7 is obtained directly from the developed Hilbert theory, without reproducing Euclid's auxiliary geometric argument.

6 Lean Formalization

The Lean proof mirrors the Hilbert reconstruction almost step by step.

The proof first derives nondegeneracy of the two triangles from the assumption that the candidate vertices lie on the same side of the given base.

A single application of the theorem `HilbertSSS` establishes congruence of the two triangles and yields equality of the corresponding base angles.

The uniqueness theorem for angle construction then shows that both candidate vertices determine the same ray from the common base vertex. Finally, the uniqueness of segment construction implies that two points lying on the same ray at the same distance from the origin must coincide.

The complete formal proof is therefore a short composition of high-level interface theorems and contains no explicit geometric construction.

7 Discussion

Proposition I.7 illustrates a fundamental difference between the historical development of Euclid's geometry and its modern axiomatic reconstruction.

In the *Elements*, Proposition I.7 serves as an essential lemma for the proof of the Side–Side–Side congruence theorem in Proposition I.8. Euclid therefore establishes the uniqueness result before proving the general congruence theorem.

The Hilbert reconstruction follows the opposite logical direction. Once the general Side–Side–Side theorem has been derived from the Hilbert axioms, Proposition I.7 becomes an immediate uniqueness corollary obtained by combining triangle congruence with the uniqueness of angle and segment constructions.

This proposition therefore demonstrates that the historical order of the *Elements* need not coincide with the logical organization of a modern formal theory. As the interface develops, results that were once foundational become concise consequences of more general synthetic theorems.